

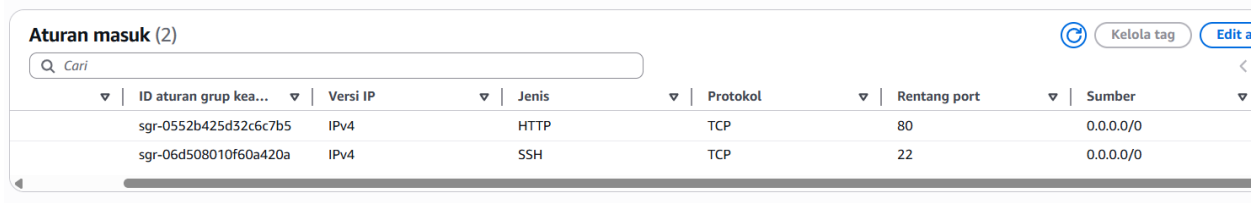
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Kelas : IF – 1

Rangkuman praktikum system terdesentralisasi dan terdistribusi

Koneksi Application Server ke Database Server MySQL Antar VM di AWS



The screenshot shows the AWS Security Groups console. At the top, it says "Aturan masuk (2)" (Inbound rules (2)). There is a search bar with the placeholder "Cari". To the right, there are buttons for "Kelola tag" (Manage tags) and "Edit a" (partially visible). Below the header, there is a table with the following columns: "ID aturan grup kea...", "Versi IP", "Jenis", "Protokol", "Rentang port", and "Sumber". The table contains two rows of inbound rules.

ID aturan grup kea...	Versi IP	Jenis	Protokol	Rentang port	Sumber
sgr-0552b425d32c6c7b5	IPv4	HTTP	TCP	80	0.0.0.0/0
sgr-06d508010f60a420a	IPv4	SSH	TCP	22	0.0.0.0/0

Pembahasan : membuat vm-server-application, dengan aturan masuk http dengan 0.0.0.0 agar bisa diakses semua IP

```

Get:18 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports/main amd64 f Metadata [116 B]
Get:41 http://security.ubuntu.com/ubuntu noble-security/main Translation-en amd64 8.3.6-0ubuntu0.24.04.5 [127 kB]
Get:42 http://security.ubuntu.com/ubuntu noble-security/main amd64 Compone amd64 8.3.6-0ubuntu0.24.04.5 [127 kB]
Get:43 http://security.ubuntu.com/ubuntu noble-security/main amd64 c-n-f Me amd64 8.3.6-0ubuntu0.24.04.5 [127 kB]
Get:44 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Pac amd64 8.3.6-0ubuntu0.24.04.5 [127 kB]
Get:45 http://security.ubuntu.com/ubuntu noble-security/universe Translati amd64 8.3.6-0ubuntu0.24.04.5 [127 kB]
Get:46 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Com amd64 8.3.6-0ubuntu0.24.04.5 [127 kB]
Get:47 http://security.ubuntu.com/ubuntu noble-security/universe amd64 c-n amd64 8.3.6-0ubuntu0.24.04.5 [127 kB]
Get:48 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Pa amd64 8.3.6-0ubuntu0.24.04.5 [127 kB]
Get:49 http://security.ubuntu.com/ubuntu noble-security/restricted Translat amd64 8.3.6-0ubuntu0.24.04.5 [127 kB]
Get:50 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Co amd64 8.3.6-0ubuntu0.24.04.5 [127 kB]
Get:51 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Pa amd64 8.3.6-0ubuntu0.24.04.5 [127 kB]
Get:52 http://security.ubuntu.com/ubuntu noble-security/multiverse Translat amd64 8.3.6-0ubuntu0.24.04.5 [127 kB]
Get:53 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Co amd64 8.3.6-0ubuntu0.24.04.5 [127 kB]
Get:54 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 c- amd64 8.3.6-0ubuntu0.24.04.5 [127 kB]
94% [19 Components-amd64 store 0 B]

```

i-0a454cf2bfe79bea1 (vm-server-application)

PublicIPs: 44.223.49.126 PrivateIPs: 172.31.66.247

```

tu2 [4076 B]
Get:18 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main am amd64 8.3.6-0ubuntu0.24.04.5 [127 kB]
Get:19 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/main amd64 php-93ubuntu2 [1838 B]
Get:20 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/main amd64 ssl-ubuntu1 [17.8 kB]
Fetched 7133 kB in 0s (65.8 MB/s)
Preconfiguring packages ...
Progress: [ 51%] [#####]
systemd/system/phpsessionclean.timer./timers.target.wants/phpsessionclean.tim

```

i-0a454cf2bfe79bea1 (vm-server-application)

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Pembahasan : Melakukan instalasi apache PHP dan Mysql client ; sudo apt update > sudo apt install apache2 php php-mysql -y > sudo apt install mysql-client-core-8.0 -y

```
ubuntu@ip-172-31-66-247:~$ cd /var/www/html
sudo git clone https://github.com/akbarwjyy/crud-students.git
Cloning into 'crud-students'...
remote: Enumerating objects: 13, done.
remote: Counting objects: 100% (13/13), done.
remote: Compressing objects: 100% (11/11), done.
remote: Total 13 (delta 3), reused 12 (delta 2), pack-reused 0 (from the origin)
Receiving objects: 100% (13/13), 8.15 KiB | 2.72 MiB/s, done.
Resolving deltas: 100% (3/3), done.
ubuntu@ip-172-31-66-247:/var/www/html$
```

i-0a454cf2bfe79bea1 (vm-server-application)

PublicIPs: 44.223.49.126 PrivateIPs: 172.31.66.247

Pembahasan : Masuk ke dalam directory /var/www/html kemudian akses repository yang disediakan ; cd /var/www/html > sudo git clone <https://github.com/akbarwjyy/crud-students.git> > cd crud-students

```
<?php
$host = "172.31.19.37";
$user = "Irvan";
$pass = "password123";
$db    = "mahasiswa";

$conn = new mysqli($host, $user, $pass, $db);
if ($conn->connect_error) {
    die("Koneksi gagal: " . $conn->connect_error);
}
```

Pembahasan : Berikutnya mengkondigurasi koneksi antara vm server application ke vm server database , dengan perintah ; sudo nano config.php ; ip diatas Adalah ip vm server database

```
ubuntu@ip-172-31-66-247:/var/www/html/crud-students$ mysql -h 172.31.19.37 -u akbar -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 12
Server version: 8.0.43-0ubuntu0.24.04.2 (Ubuntu)

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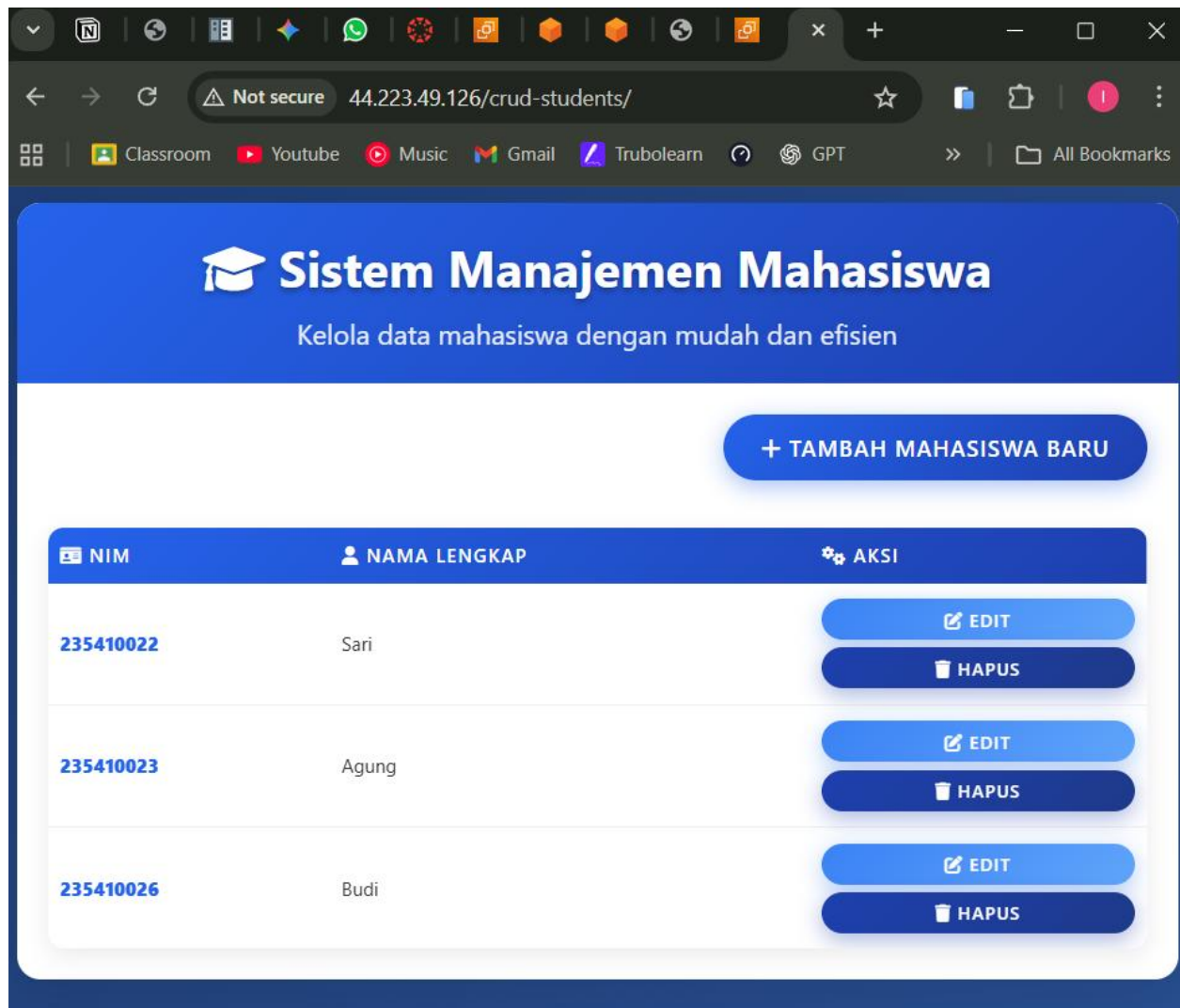
Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> █
```

i-0a454cf2bfe79bea1 (vm-server-application) 

PublicIPs: 44.223.49.126 PrivateIPs: 172.31.66.247

Pembahasan : Menguji koneksi dengan perintah ; mysql -h 172.31.19.37 -u akbar -p, jika muncul diatas maka berhasil terhubung antar app dan database vm nya



Pembahasan : Yang terakhir kita membuka browser dengan link <http://44.223.49.126/crud-students/index.php> public ip sesuaikan pakai vm server application maka akan terbuka tampilan diatas

Kesimpulan :